

Oogenesis and ovarian follicles

What happens to the oocyte and its surrounding cells before and after ovulation?

Learning goal

Distinguish an oocyte, a follicle and a corpus luteum, and explain how their roles change across ovulation.

Before you begin

An ovary is an organ. An oocyte is a cell within it. Do not use these names interchangeably.

How to complete this lesson Learn → worked example → practise → lesson check

- Read the explanation and its diagrams. Open a bold term when you need a definition.
- Follow the worked example, then try both supported questions. Select Check answer for feedback.
- Open the required check and select Start. Correct both selections to unlock writing.
- Write each explanation and select its Save written response button. Editing saved writing makes it a draft again.
- Select Submit and finish to complete the check. Find your records in All My Work.

Expected work: one completed lesson check. Supported practice, videos and textbook questions are optional. Saved writing is not automatically graded.

Chapter 14 · printed pp. 482–497

Open p. 482 · Open p. 484 · Open p. 496 · Open p. 497

Key terms: oogenesis · oocyte · follicle · corpus luteum

Vocabulary help

Key terms

Words for this lesson

Use these definitions when you need help with a term.

oogenesis

The process producing the female gamete.

oocyte

A developing female reproductive cell.

follicle

An ovarian structure containing an oocyte and supporting cells.

corpus luteum

A temporary endocrine structure formed from follicular tissue after ovulation.

More vocabulary for this lesson

Lesson vocabulary

polar body

A small cell receiving relatively little cytoplasm during oogenesis.

An oocyte develops inside a larger supporting structure

An ovary, an oocyte and a follicle are related, but they are not three names for the same thing. The ovary is the organ. An oocyte is a developing female reproductive cell within it. A follicle includes that oocyte and the supporting cells around it. Oogenesis is the process that produces the female gamete.

Many oocytes begin development before birth and enter meiosis, the chromosome-reduction process. They pause during meiosis I for an extended period. From puberty onward, hormonal signals support the later development of groups of follicles. Usually one becomes the dominant follicle that proceeds toward ovulation in a cycle.

The ovaries begin life with a finite supply of oocytes rather than continually replenishing them as spermatogonia maintain the male starting-cell supply. Many follicles do not reach ovulation. A drawing with several stages is not a count of the eggs that will be released in one cycle.

The two ovaries do not have to take perfectly alternating turns. The textbook's simplified monthly description should not be used to infer which ovary must release the next oocyte. Nor does a usual one-oocyte example mean that every person's cycle follows the same timetable.

Chromosome reduction and cytoplasm division are different questions

Cytoplasm is the cell material outside the nucleus, including resources and organelles that support cell activity. Cytokinesis is the division of that cytoplasm when cells separate. In oogenesis, cytokinesis is unequal: most of the cytoplasm stays in one large reproductive cell, while small polar bodies receive much less.

This unequal division concentrates resources in the large cell that can contribute to early development after fertilization. It does not give that cell an extra chromosome set. Haploid means one set of chromosomes, normally 23 in humans. Diploid means two sets, normally 46 chromosomes in a human body cell. Cell size, amount of cytoplasm and chromosome-set number describe different properties.

Before ovulation, a primary oocyte completes meiosis I, producing a large secondary oocyte and a small polar body. The secondary oocyte begins meiosis II and pauses again. It is the cell released at ovulation: it has one chromosome set, but has not yet finished separating the copied chromosome material in meiosis II. A sperm's entry can trigger completion of that division.

Compare the functional products. In spermatogenesis, the process of sperm production in the testes, one primary spermatocyte can produce four haploid spermatids that differentiate into sperm. Oogenesis concentrates most cytoplasm in one large reproductive cell, with small polar bodies rather than four equal-sized functional eggs. Both processes reduce chromosome sets. Their distribution of cytoplasm and timing are different.

The familiar word egg and the figure label ovum are often used broadly. Use secondary oocyte when you need the precise stage released at ovulation. It is not necessary to memorise every meiotic phase here, but it is important not to say that meiosis II was already finished before release.

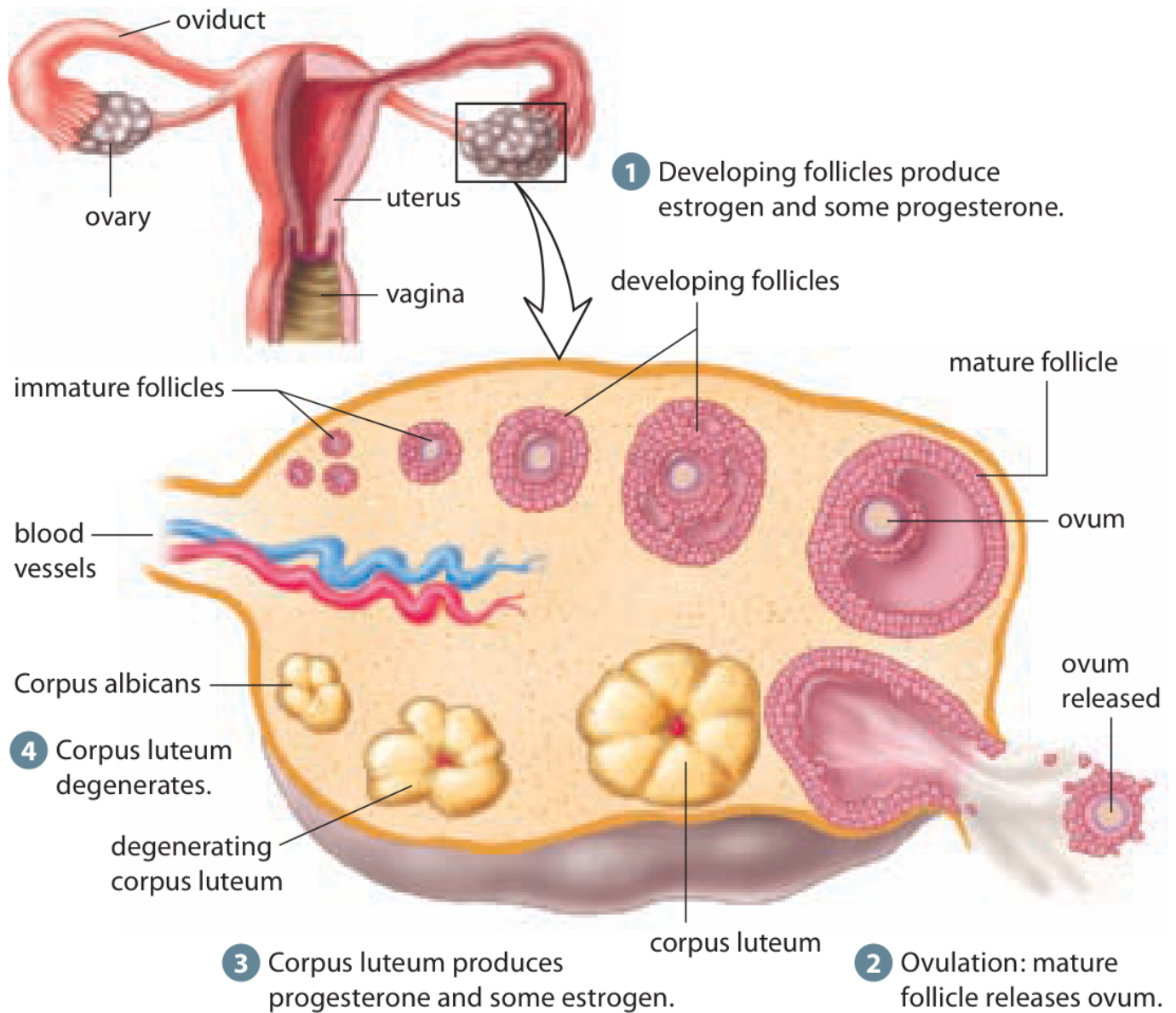
Read the follicle diagram as a sequence of changes

As a follicle develops, its supporting cells grow and its estrogen output increases. The tissue therefore supports a reproductive cell and also acts as an endocrine source, releasing hormones into the blood. Estrogen helps stimulate growth of the endometrium in the uterus. The ovarian source and uterine target are in different places.

Begin with the small overview at the top of the figure. The box and broad arrow enlarge an ovary; they do not show an ovary moving. In the large view, find the small immature follicles toward the upper left, larger developing follicles across the top and the mature follicle at the right. Locate the cell labelled ovum within the mature follicle rather than treating the whole fluid-filled structure as the egg.

At the lower right, the mature follicle ruptures and the oocyte is released. This is ovulation. Most follicular tissue remains within the ovary. The released cell and the remaining supporting tissue now have different destinations and functions.

The sequence is spread around the drawn ovary so the stages fit in one picture. A follicle does not physically travel clockwise around the organ to become a corpus luteum. The illustration combines stages through time, not necessarily all the structures visible together at one instant. Read the changes in the tissue, rather than treating its drawing position as a migration route.



An oocyte leaves the mature follicle; hormone-producing tissue remains in the ovary. Inquiry into Biology, p. 496.

Hormone-producing tissue remains after the oocyte leaves

The follicular tissue left after ovulation develops into the corpus luteum, a temporary endocrine structure within the ovary. It releases mainly progesterone and some estrogen. Progesterone helps maintain the prepared endometrium after ovulation. The corpus luteum remains in the ovary; its hormones enter blood and reach target tissues, including the uterus.

Follow the figure from the released cell at the lower right to the yellow corpus luteum lower in the ovary. The cell outside the ovary is not the yellow hormone-producing structure. Next find the smaller degenerating corpus luteum at the lower left. The labelled corpus albicans is the scar-like remnant after regression, not an embryo.

If the corpus luteum does not receive a pregnancy-supporting signal, it regresses and its hormone output falls. The resulting loss of hormone support contributes to shedding of part of the endometrium during menstruation. This occurs later in the sequence; the death of an unfertilized oocyte does not instantly cause menstrual flow the next day. The timing and

interacting feedback will be developed with the ovarian and uterine cycles.

Keep both paths in one explanation: the oocyte is released toward the oviduct, while remaining ovarian tissue changes its hormone output. A cell leaves, but the organ can continue sending a chemical signal. This makes it possible for events in the ovary to prepare and maintain tissue somewhere else.

Worked example

A released cell and an ovarian signal can exist at the same time

A model observation places a recently released oocyte in an oviduct and shows an active corpus luteum in the ovary. A student says the progesterone source must have travelled out with the oocyte. Does the observation support that claim?

- Identify the two structures in the observation. The oocyte is the released reproductive cell. The corpus luteum is tissue formed from the follicle remaining in the ovary.
- Identify the hormone source. The active corpus luteum releases progesterone and some estrogen while it is still in the ovary. The model therefore already provides an ovarian source; it does not require the released cell to carry that tissue with it.
- Follow the hormone route. Hormones enter the blood and can act on the endometrium in the uterus. The movement of a signal through blood is different from the cell's movement through an oviduct.
- Check the conclusion. Release of the oocyte is compatible with continuing ovarian hormone output. Neither the corpus luteum nor the whole ovary needs to travel to the uterus. The observation alone does not establish that fertilization or implantation has occurred.

Optional application

In a stated model, a small polar body and a much larger secondary oocyte each have one haploid chromosome set. A student says the larger cell must be diploid because it contains more material. Explain the mistake and what the unequal size does help preserve.

Answer withheld for first attempt

Practise with support

Apply the explanation

Try both questions. Use the hint to revise your thinking after an incorrect answer.

What does a follicle contain before ovulation?

- A complete embryo implanted in the uterus
- An oocyte and supporting cells
- Only a mature sperm

Which structure forms from tissue remaining after the oocyte is released?

- Endometrium
- Epididymis
- Corpus luteum

Stop and think

Try this before opening the explanation

Does the corpus luteum travel to the uterus?

Answer withheld for first attempt

Open required check · Oogenesis and ovarian follicles

Explain what you have learned

This check counts toward chapter progress. Correct the 2 selections to unlock 2 written explanations. Saving writing records your response; it does not grade its accuracy.

Selection 1

Which statement correctly distinguishes an oocyte from the corpus luteum?

- The oocyte forms from the uterine lining
- The corpus luteum is the released oocyte
- The oocyte is released; the corpus luteum forms from tissue remaining in the ovary
- Both move through the oviduct together

Selection 2

Why is cytokinesis unequal during oogenesis?

- It doubles the final chromosome number
- It creates four equally sized motile gametes
- It prevents any cell from becoming haploid
- It retains most cytoplasm in one large reproductive cell

Answer the selections correctly to unlock the writing.

Written explanation 1

Follow an ovarian follicle from growth through ovulation and corpus-luteum formation. Include the changing hormone output.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

Written explanation 2

Compare the products of spermatogenesis and oogenesis without giving the detailed phases of meiosis.

[Write your own response separately.]

Maximum 3200 characters. Explain the biology in your own words.

Answer withheld for first attempt

← Female reproductive structures and the egg pathway / Next: Coordinating the ovarian and uterine cycles →